

Project Presented by:

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LICENSE PLATE DETECTION

PROBLEM STATEMENT



- PROBLEM:
 - Manual plate logging during security rounds is slow,, and tedious
- Stakeholders:
 - Commercial property managers and security teams
- Impact:
 - Wastes patrol time, introduces human error, and delays spotting unauthorized vehicles
- Solution
 - Automated License Plate Detection for digital database cross-checks.

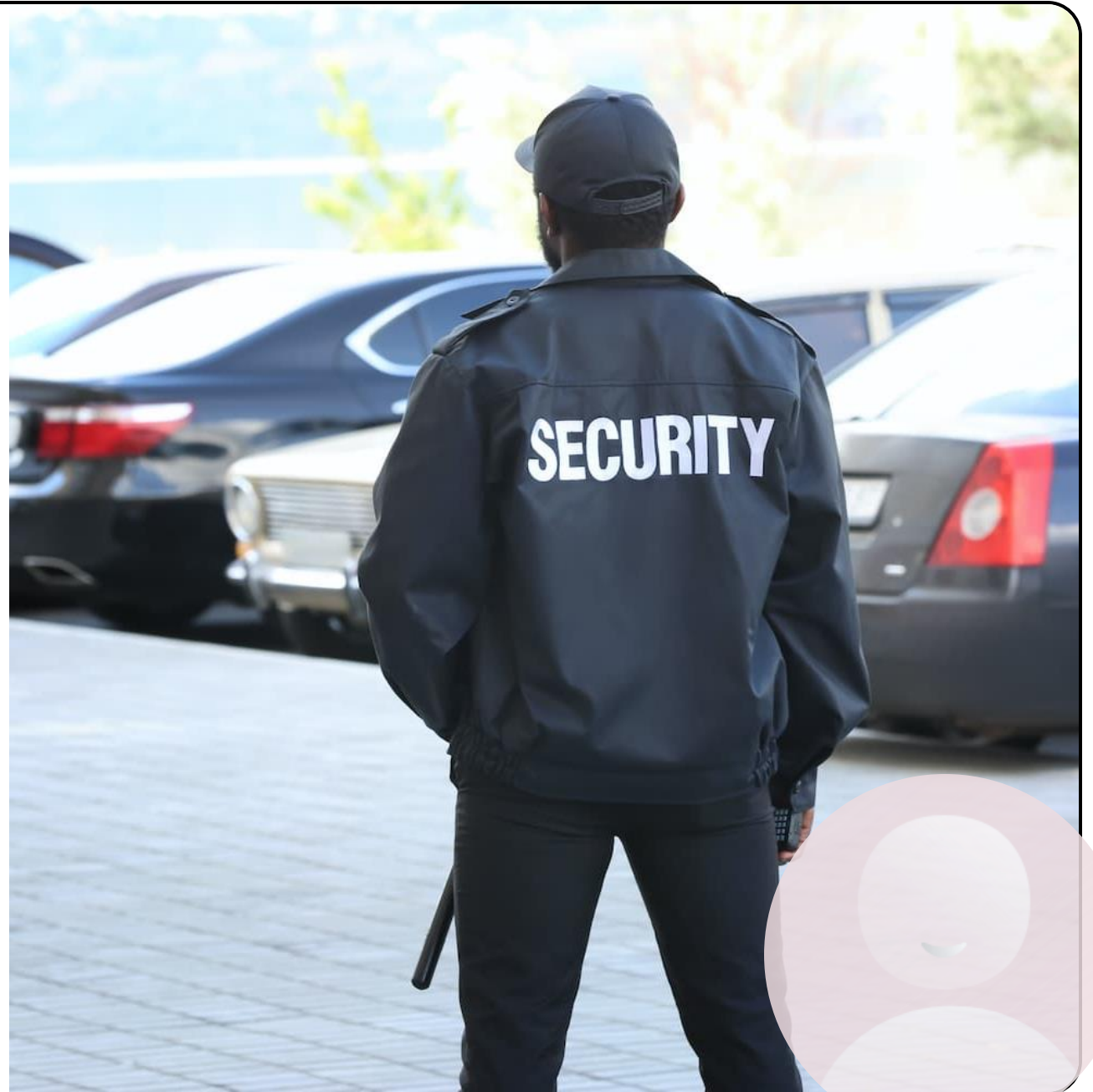
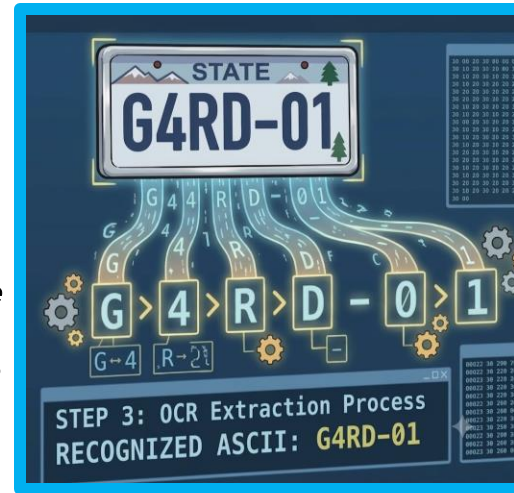




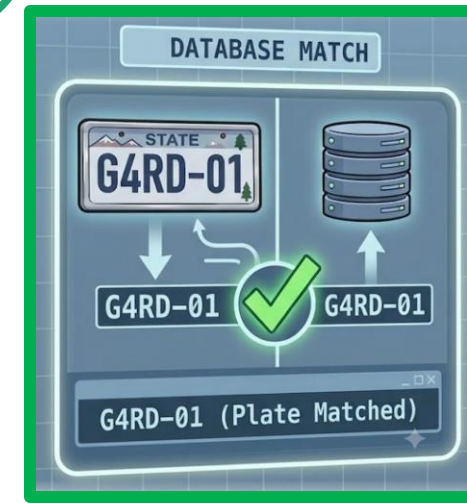
PLATE DETECT

YOLO11 locates license plate with bounding box & confidence score



DB MATCH

Database Cross-Check & Alert: final output & actionable intelligence



CAMERA INPUT

Captures vehicle image for license plate processing

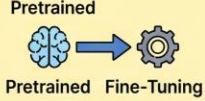


OCR EXTRACT

Converts License Plate to Text w/EasyOCR:
Tier 2 upgrade

SOLUTION ROADMAP

CV Technical Approach

	 Required Item	 Selection	 Explanation
1	 CV Technique	 Object Detection	The model locates license plates within parking lot images by drawing bounding boxes around detected plates. 
2	 Model Architecture	 YOLO11n (You Only Look Once - Nano)	A lightweight, single-stage detection architecture designed for fast and efficient image analysis. 
3	 Model Usage	 Transfer Learning (Fine-Tuning)	Starts with a pretrained YOLO model and adapts it to recognize license platet using a labeled dataset. 
4	 Framework	 PyTorch + Ultralytics YOLO	PyTorch provides the deep learning foundation, while Ultralytics provides the YOLO implementation for training and inference. 

- Project will utilize YOLO11n and transfer learning to prioritize high-speed, real-time detection without sacrificing accuracy.
- The **Ultralytics** framework provides a robust foundation for training a model that is lightweight enough for edge-case access control applications

Data Acquisition Strategy



Source & Provenance

Public dataset from Roboflow Universe (Project: *rxg4e*, Version 13).



Scale & Volume

Total of 10,125 high-quality images (70% Train, 20% Valid, 10% Test).



Label Specification

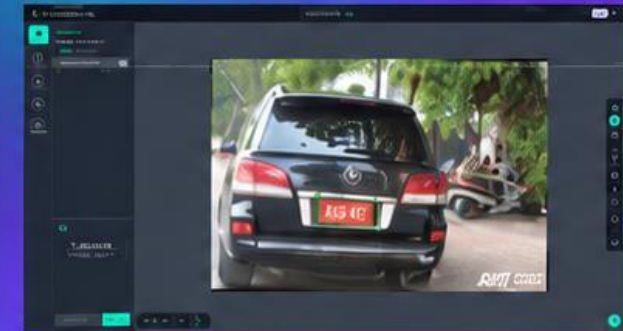
Annotated with precise bounding boxes for the class: **license-plate**.



Preparation Status

Pre-labeled and cleaned; ready for immediate YOLO training and deployment.

How To Train and Deploy an ANPR System

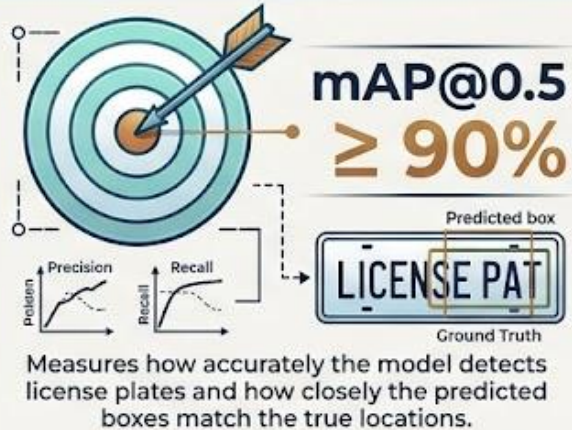


roboflow

LICENSE PLATE DETECTION (YOLOv11)

PROJECT SUCCESS METRICS

PRIMARY METRIC



SECONDARY METRIC



EVALUATION STEPS

Stage 1

- Train on labeled training images



Stage 2

- Measure mAP on labeled test images



Stage 3

- Demonstrate detections on new images from a parking lot



Data sources

Future extensions: OCR (EasyOCR)

Success Metrics

Phase	Goal	Milestone	10-week Term
BluePrint -- LPR	Plan approval	Midterm submission (plan)	07/13
First Working Demo	Proof of Concept for LPR work	Minimally Viable Product (DEV)	07/17
Make It Yours	Add My Parking Lot Data	QA on Functional Project: UAT/SIT	07/22
Improve/Measure	Test/refine Final project	Metrics Recorded (PROD)	07/27
Package & Present	Demo, README.md, slides	Final Submission for business analysis (hypercarE)	07/29

MILESTONE PLAN—License Plate Recognition CV Project

RISKS and resources (A)



Risk	Probability	Mitigation Plan
Model performance drops under varied lighting or viewing angles	<i>Low</i>	Apply image augmentation (brightness, rotation, blur) or retrain using a more diverse dataset.
EasyOCR misreads license plate characters due to low-resolution or blurred crops	<i>High</i>	Preprocess cropped plates using upscaling, grayscale conversion, and contrast enhancement before OCR.

Technology Stack Overview

Risks and **RESOURCES** (B)

LAYER	TECHNOLOGY	PURPOSE IN PROJECT
 Programming	Python	Core language for AI and computer vision development.
 Environment	Google Colab	Free GPU-accelerated cloud environment for training.
 Object Detection	Ultralytics YOLO	Detects plates and generates high-precision bounding boxes.
 Image Processing	OpenCV	Loading images, preprocessing, and visual annotation.
 Dataset Source	Roboflow / Kaggle	Provides labeled license plate images for model training.

Expected Cost: \$0.00